

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve Any Two of the following.

(12)

A) Draw the following lines according to drawing standard SP 46.

Remember

6

1. Projection Line; 2. Center Line

B) Draw a regular hexagon of 25 mm side.

Remember

6

C) Explain Unidirectional system of dimensioning by drawing suitable diagrams.

Understand

6

Q.2 Solve Any one of the following.

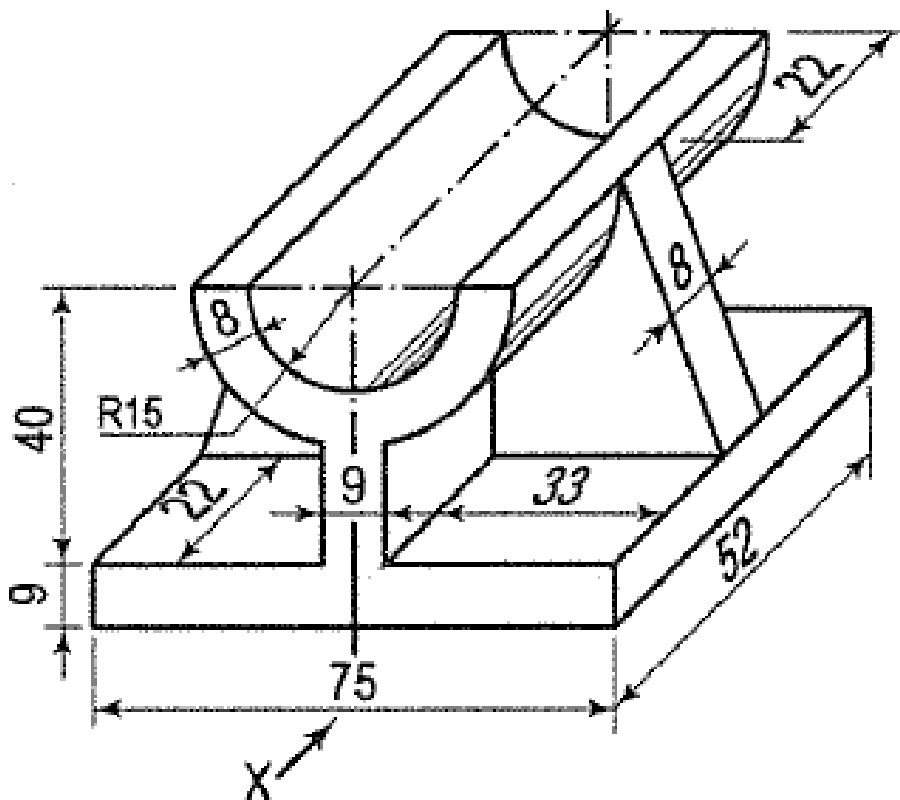
(12)

A) Following figure shows the pictorial view of an object, draw its front view

12

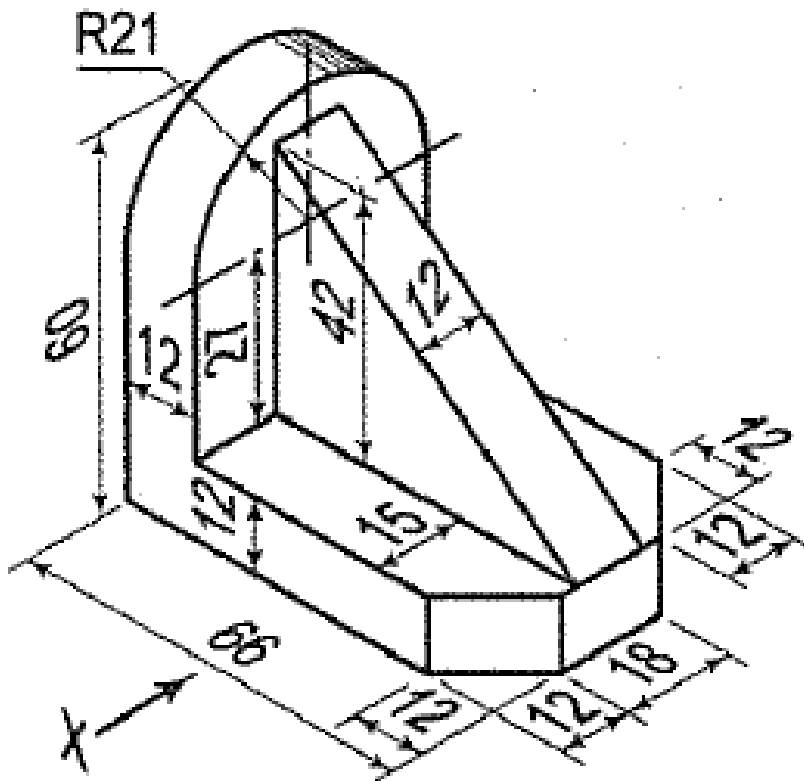
(FV) looking from direction of X, and the top view (TV) by using First angle Projection method.

Apply



- B)** Following figure shows the pictorial view of an object, draw its front view (FV) looking from direction of X, top view (TV) and right hand side view. Symbolically mention the projection method used.

Apply **12**



Q. 3 Solve Any One of the following.

(12)

- A)** A line AB, 65 mm long has its end A 20 mm above the H.P. and 25 mm in front of the V.P. The end B is 40 mm above the H.P. and 65 mm in front of the V.P. Draw the projections of AB and show its inclinations with the H.P. and the V.P.
- B)** Draw the projections of a regular hexagon of 25 mm side, having one of its sides in the H.P. and inclined at 60° to the V.P., and its surface making an angle of 45° with the H.P.

Evaluate **12**

Evaluate **12**

Q.4 Solve Any One of the following.

(12)

- A)** A pentagonal pyramid base 25 mm side and axis 50 mm long has one of its triangular faces in the V.P. and edge of the base contained by that face makes an angle of 30° with the H.P. Draw its projections.
- B)** A pentagonal prism is resting on one of the corners of its base on the H.P. The longer edge containing that corner is inclined at 45° to the H.P. The top view of the axis of the prism makes an angle of 30° to the V.P. Draw the projections of the solid.

Evaluate **12**

Evaluate **12**

Q. 5 Solve Any one of the following.

(12)

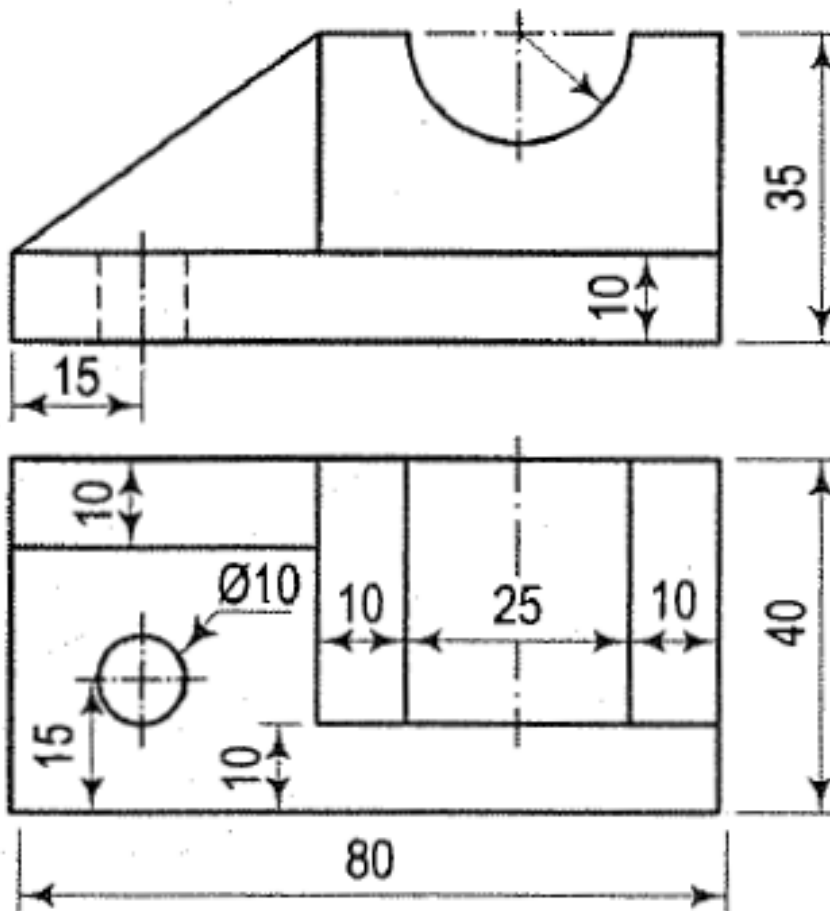
- A)** A hexagonal pyramid, base 30 mm side and axis 65 mm long, is resting on its base on the H.P. with two edges parallel to the V.P. It is cut by a section plane, perpendicular to the V.P. inclined at 45° to the H.P. and intersecting the axis at a point 25 mm above the base. Draw the front view, sectional top view, sectional side view and true shape of the section.
- B)** Draw the isometric view of the following object having FV and TV drawn by first angle projection method.

Evaluate

12

Synthesize

12



*** End ***